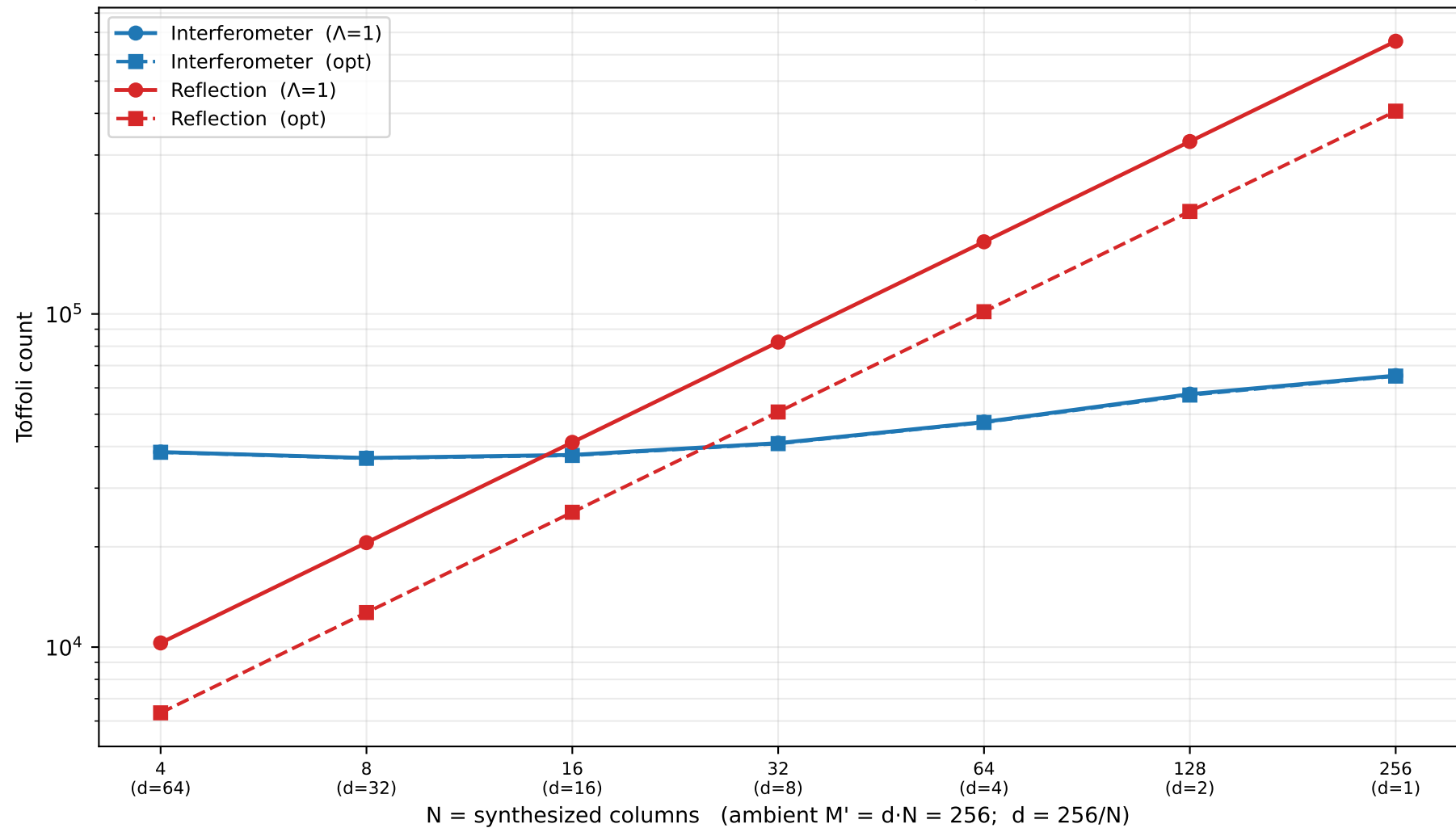
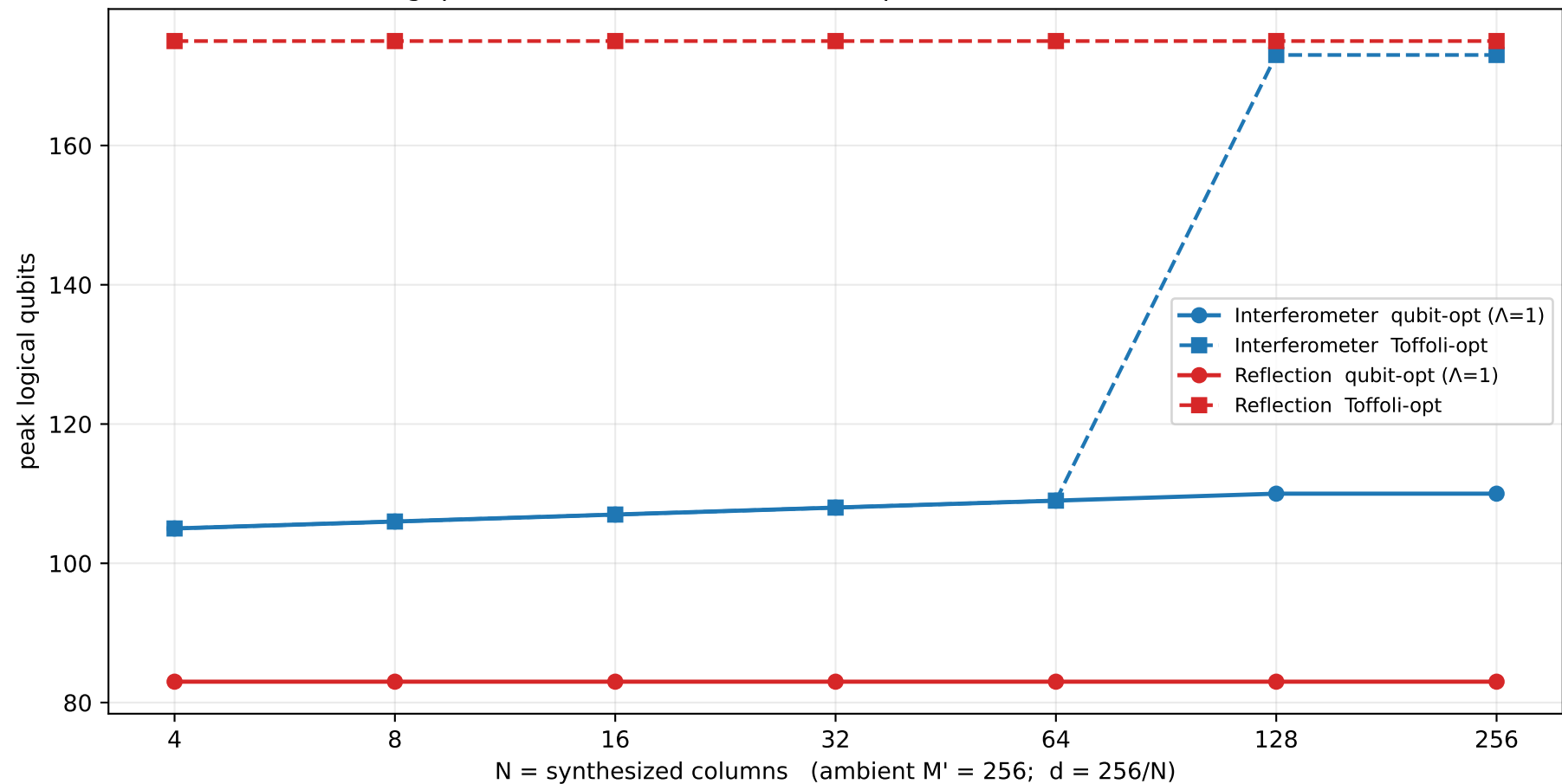


Isometry synthesis at fixed ambient $M' = 256$: Toffoli vs N
(interferometer vs reflection, $b=32$, single block)



Tradeoff at fixed ambient $M' = 256$, $b = 32$: Toffoli-optimal spends qubits
(gap between dashed and solid = qubits added to lower Toffoli)



Isometry synthesis Toffoli/qubit tradeoff at fixed ambient $M' = d \cdot N = 256$ ($b=32$)
 $\Lambda=1$ = qubit-optimal; opt = Toffoli-optimal (QROAM batching)

N	d	Inter T($\Lambda=1$)	Inter q($\Lambda=1$)	Inter T(opt)	Inter q(opt)	Refle T($\Lambda=1$)	Refle q($\Lambda=1$)	Refle T(opt)	Refle q(opt)
4	64	38,506	105	38,443	105	10,292	83	6,348	175
8	32	36,962	106	36,869	106	20,584	83	12,696	175
16	16	37,746	107	37,611	107	41,168	83	25,392	175
32	8	40,954	108	40,807	108	82,336	83	50,784	175
64	4	47,390	109	47,243	109	164,672	83	101,568	175
128	2	57,449	110	57,089	173	329,344	83	203,136	175
256	1	65,305	110	65,082	173	658,688	83	406,272	175